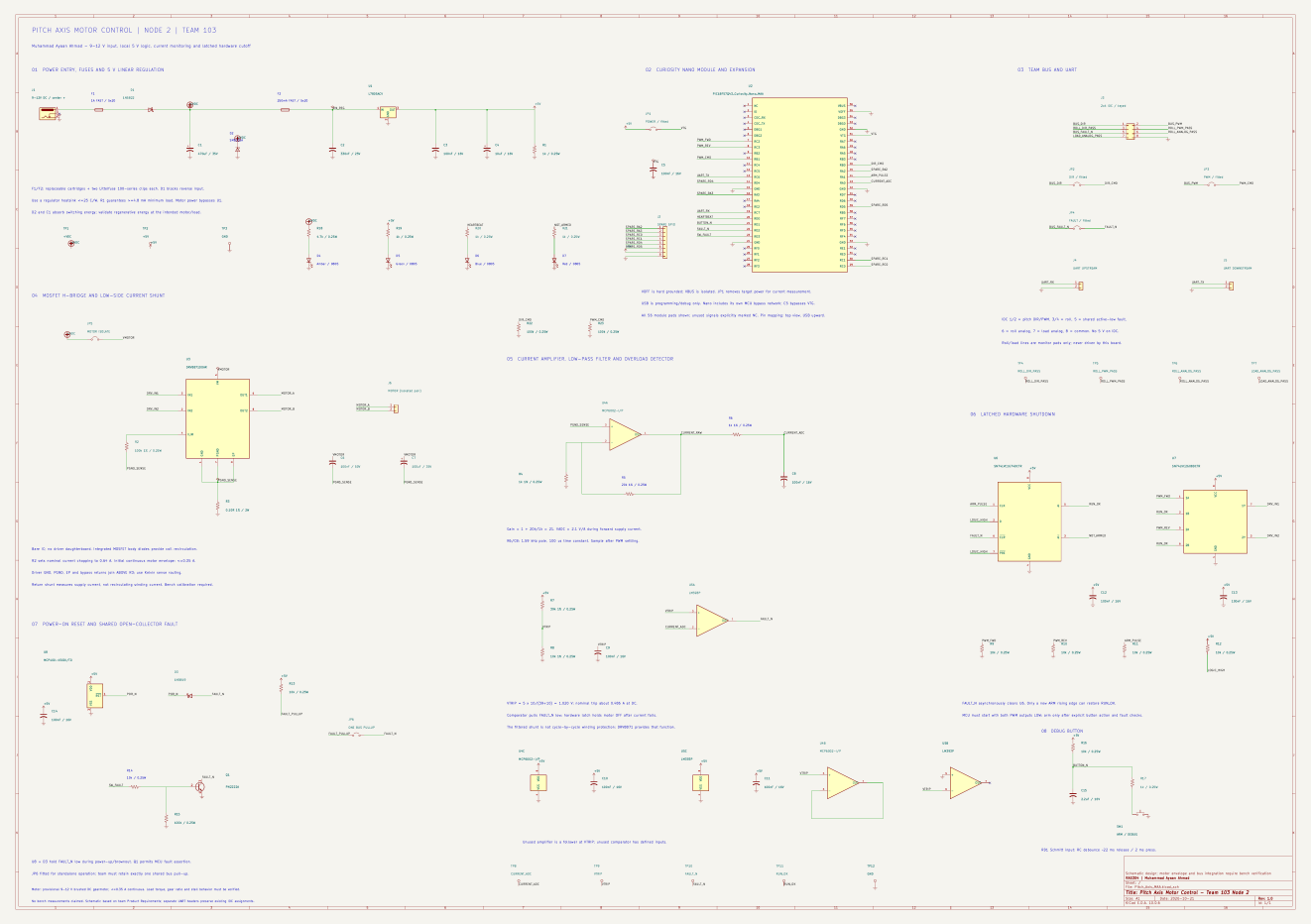
Schematics: Pitch-axis motor control

Muhammad Ayaan Ahmad | RAS304 Team 103 | Node 2 | Revision 1.0 | 21 October 2026

Schematic captured and checked. Bench testing, motor selection and team UART integration remain open.

Individual Schematics page | Vector schematic | KiCad project ZIP



Full vector schematic follows as the final appendix page. Zoom to inspect component values.

Design basis and scope

This schematic implements Node 2, the pitch-axis actuation board assigned to Ayaan Ahmad in Team 103's published Product Requirements. It contains the required Curiosity Nano module, local regulated power, a bare MOSFET H-bridge IC, low-side current sensing, an op-amp, filtering, overload detection, and debug interfaces.

The earlier individual component-selection page described a provisional optical sensor. This schematic follows the explicit team allocation instead. The team block-diagram page is still a template. UART routing is therefore provided on separate upstream/downstream headers while the published IDC assignments are preserved.

This is a schematic-stage design. No breadboard tests, thermal tests, mechanical performance, or measured fault-response times are claimed. The provisional motor envelope must be checked against the actual motor and gearbox before PCB release.

Power and protection

J1 accepts a regulated 9-12 V, center-positive barrel supply. Select a source rated at least 2 A. F1 is a removable 1 A fast 5x20 mm fuse; D1 is a 1N5822 reverse-polarity blocking diode. C1 (470 uF/35 V) and D2 (SMBJ15A) support transient suppression. Actual motor regeneration energy and fuse inrush coordination require bench verification.

F2 independently protects the logic branch with a removable 250 mA fast fuse. U1 is the tighter-tolerance L7805ACV through-hole regulator, with 330 nF input bypass, 100 nF output bypass, 10 uF bulk output capacitance and a 1 kohm minimum-load resistor. All listed capacitor voltages exceed operating voltage by at least 25%.

A 150 mA maximum logic-rail design allocation gives a conservative regulator loss of (12-4.8) x 0.150 + 12 x 0.010 = 1.20 W, ignoring the helpful D1 drop. A heatsink <=25 C/W, with about 5 C/W junction-to-case and 2 C/W interface, estimates <=38.4 C rise. This is a design allocation and screening calculation, not a validated maximum-current budget or enclosure test.

The Nano receives local 5 V at VTG through JP1. VOFF is tied to common to disable its on-board target regulator; VBUS is explicitly unconnected. C5 bypasses VTG. The module already contains the MCU supply bypass network. USB remains available for programming and debugging.

Motor drive and current sensing

U3 is a DRV8871DDAR mounted directly on the PCB. It integrates the MOSFET H-bridge and inductive recirculation paths. JP5 isolates the motor branch. J6 carries the two motor terminals; neither terminal is tied to ground. The initial motor envelope is a 9-12 V brushed DC gearmotor at <=0.35 A continuous. Torque, ratio, speed, startup current, travel sensing and actual motor part number remain to be established.

R2 = 100 kohm sets nominal internal current chopping at 64/100 = 0.64 A. Applying the datasheet 59-69 kV coefficient range and 1% resistor tolerance gives approximately 0.584-0.697 A at the published test conditions. This is not a guaranteed characterization at every load and temperature.

R3 = 0.10 ohm/2 W sits below the entire driver return island. U3 GND, PGND, exposed pad, ILIM return, and local bypass returns all meet above this resistor. Kelvin sense wiring feeds U4A. Do not bypass the shunt with a ground plane or test lead. The measured signal is net supply-return current; it excludes winding current circulating wholly inside the bridge.

U4A (MCP6002) has non-inverting gain 1 + 20k/1k = 21, giving 2.1 V/A for forward supply current. R6 = 1 kohm and C8 = 100 nF form a low-pass stage with fc = 1/(2 pi RC) = 1.59 kHz and time constant 100 us. RA0 reads the filtered signal. ADC timing must account for PWM, amplifier settling, source impedance and the required acquisition time.

R7/R8 produce VTRIP = 5 x 10/(39+10) = 1.020 V. U5A pulls FAULT_N low when filtered current exceeds this reference, corresponding to about 0.486 A at steady DC. Supply tolerance, resistor tolerance, amplifier/comparator offset and PWM duty affect the actual trip point. At 0.697 A, R3 dissipates about 0.049 W, well below its 2 W rating.

Latched shutdown and debugging

U6, an SN74LVC1G74, latches the motor permit. FAULT_N asynchronously clears it; U7 gates both direction/PWM outputs to zero when the permit is cleared. The driver therefore coasts after a fault and remains disabled even after current falls. A fresh ARM_PULSE edge is required to restore permission. This is a functional interlock, not a certified safety controller.

U8 (MCP100-450) and D3 assert the common fault during power-up or brownout. Q1 lets the MCU assert the same open-collector fault. JP6 installs the local 10 kohm pull-up for standalone tests; use exactly one pull-up across the connected bus. JP4 separates the local fault from IDC pin 5 for debugging.

SW1 uses a 10 kohm pull-up, 2.2 uF capacitor and 1 kohm discharge resistor. The approximate RC times are 22 ms release and 2 ms press. Configure RD1 for Schmitt-trigger digital input and verify actual switch bounce. The button requests arming; firmware must first check faults and require a deliberate action.

D4 indicates the input rail, D5 indicates 5 V, D6 is an MCU heartbeat, and D7 shows not-armed status. Each has a current-limiting resistor. Twelve test points and an eight-pin expansion header expose key signals and spare GPIO. Unused op-amp and comparator inputs have defined states. All IC supply pins have local bypassing.

Verification and next-stage work

KiCad 10.0.6 electrical-rule check: 0 errors and 0 warnings, no exclusions. An independent check of the exported netlist verified 17 critical connection groups, VBUS/VTG isolation, and the shunt return separation. Custom module, driver, amplifier, comparator and logic symbols include MAA in their library names. All schematic footprints are packaged locally.

Before manufacturing: confirm the physical motor and gearbox; validate driver thermal copper/exposed-pad construction, startup and stall behavior, regenerative voltage, fuse curves, regulator heating and the overload threshold. Confirm the upstream/downstream UART connections and the single fault pull-up with the team. Obtain instructor feedback on filter choice.

Before integration: perform logic-only power-up with JP5 removed; measure 5 V and verify the latch starts cleared. Check the driver truth table without a motor. Inject current-sense voltage or use a controlled load to exercise the trip. Confirm a fault forces both inputs low and cannot automatically restart. Then test at a current-limited supply and measure the <20 ms team fault-response target. No such measurements have been performed yet.

Team IDC interface

IDC pin	Team signal	This board
1	Pitch direction	RB0 input via JP2
2	Pitch PWM / enable	RB1 input via JP3
3	Roll direction	Test pad only
4	Roll PWM	Test pad only
5	Global FAULT_N	Open collector + RD2 + latch, via JP4
6	Roll position analog	Test pad only
7	Joint load analog	Test pad only
8	Common ground	Circuit common

PIC and module mapping

PIC signal	Module pad	Role
RA0	43	Filtered current ADC
RA1	44	ARM_PULSE
RC2 / RC3	7 / 8	PWM_FWD / PWM_REV
RC7 / RC6	19 / 13	UART RX / TX
RD0 / RD1	20 / 21	Heartbeat / debounced button
RD2 / RD3	22 / 23	FAULT_N input / software fault output

Sources

Assignment

Team Product Requirements

Team Block Diagram (template at review)

Curiosity Nano user guide

DRV8871

MCP6002

LM393

SN74LVC1G74

SN74LVC2G08

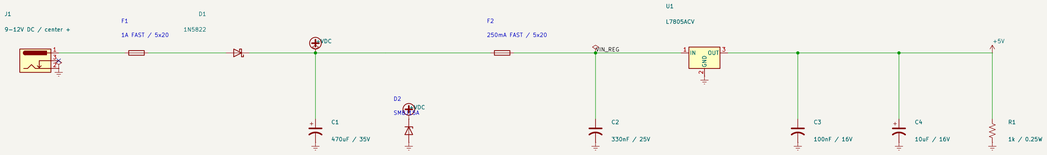
MCP100

L78 family

PITCH AXIS MOTOR CONTROL | NODE 2 | TEAM 103

Muhammad Ayaan Ahmad — 9–12 V input, local 5 V logic, current monitoring and latched hardware cutoff

01 POWER ENTRY, FUSES AND 5 V LINEAR REGULATION



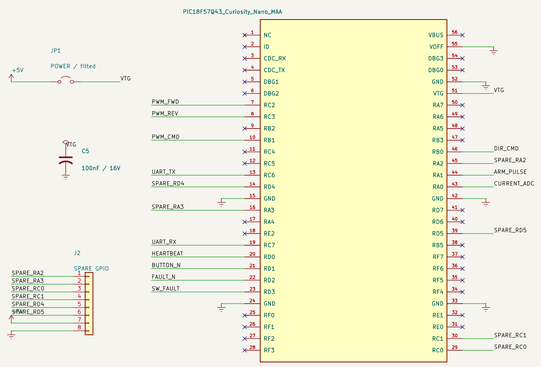
F1/F2: replaceable cartridges + two Littelfuse 100-series clips each. D1 blocks reverse input.

Use a regulator heatsink <=25 C/W. R1 guarantees >=4.8 mA minimum load. Motor power bypasses U1.

D2 and C1 absorb switching energy; validate regenerative energy at the intended motor/load.



02 CURIOSITY NANO MODULE AND EXPANSION

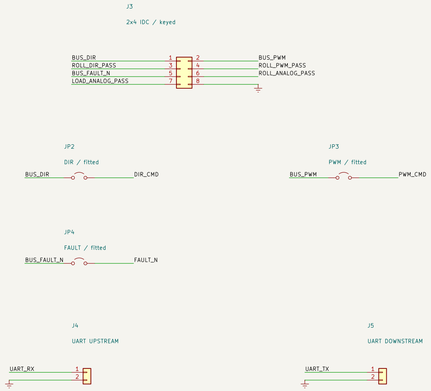


VOFF is hard grounded; VBUS is isolated. JP1 removes target power for current measurement.

USB is programming/debug only. Nano includes its own MCU bypass network; C5 bypasses VTG.

All 56 module pads shown; unused signals explicitly marked NC. Pin mapping: top view, USB upward.

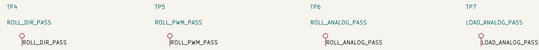
03 TEAM BUS AND UART



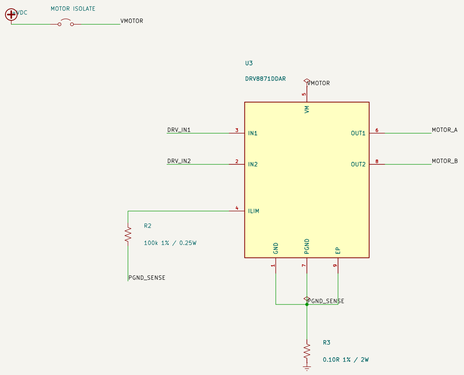
IDC 1/2 = pitch DIR/PWM, 3/4 = roll, 5 = shared active-low fault,

6 = roll analog, 7 = load analog, 8 = common. No 5 V on IDC.

Roll/load lines are monitor pads only; never driven by this board.



04 MOSFET H-BRIDGE AND LOW-SIDE CURRENT SHUNT



Bare IC; no driver daughterboard. Integrated MOSFET body diodes provide coil recirculation.

R2 sets nominal current chopping to 0.64 A. Initial continuous motor envelope: <=0.35 A.

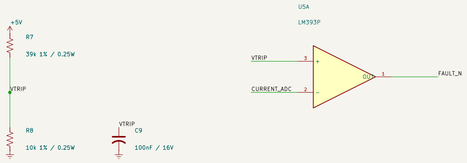
Driver GND, PGND, EP and bypass returns join ABOVE R3; use Kelvin sense routing.

Return shunt measures supply current, not recirculating winding current. Bench calibration required.

05 CURRENT AMPLIFIER, LOW-PASS FILTER AND OVERLOAD DETECTOR

Gain = 1 + 20k/1k = 21. VADC = 2.1 V/A during forward supply current.

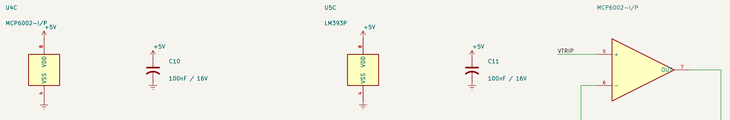
R6/C8: 1.59 kHz pole, 100 us time constant. Sample after PWM settling.



VTRIP = 5 x 10/(39+10) = 1.020 V; nominal trip about 0.486 A at DC.

Comparator pulls FAULT_N low; hardware latch holds motor OFF after current falls.

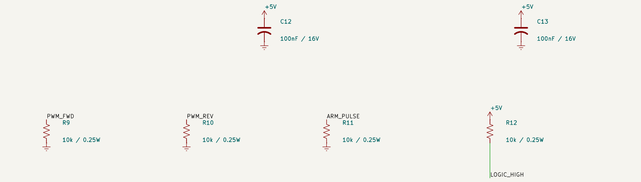
The filtered shunt is not cycle-by-cycle winding protection; DRV8871 provides that function.



Unused amplifier is a follower at VTRIP; unused comparator has defined inputs.

06 LATCHED HARDWARE SHUTDOWN

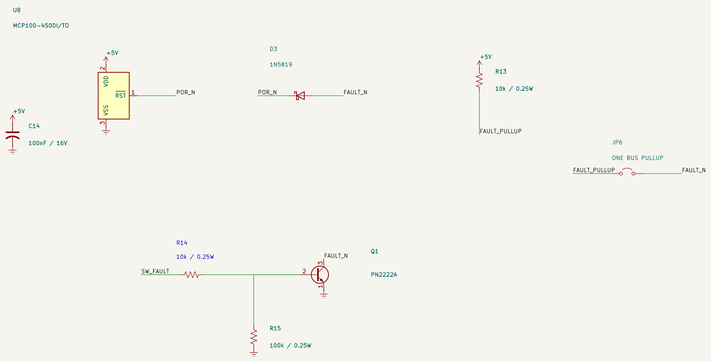




FAULT_N asynchronously clears U6. Only a new ARM rising edge can restore RUN_OK.

MCU must start with both PWM outputs LOW; arm only after explicit button action and fault checks.

07 POWER-ON RESET AND SHARED OPEN-COLLECTOR FAULT



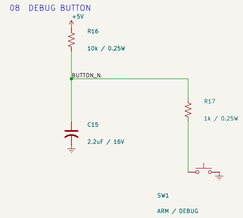
U8 + D3 hold FAULT_N low during power-up/brownout. Q1 permits MCU fault assertion.

JP6 fitted for standalone operation; team must retain exactly one shared bus pull-up.

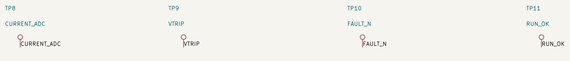
Motor: provisional 9–12 V brushed DC gearmotor, <=0.35 A continuous. Load torque, gear ratio and stall behavior must be verified.

No bench measurements claimed. Schematic based on team Product Requirements; separate UART headers preserve existing IDC assignments.

08 DEBUG BUTTON



RD1 Schmitt input: RC debounce ~22 ms release / 2 ms press.



Schematic design: motor envelope and bus integration require bench verification

RAS304 | Muhammad Ayaan Ahmad

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Title: Pitch Axis Motor Control — Team 103 Node 2

Size: A1 | Date: 2026-10-21 | **Rev: 1.0**

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